



Information Technology Institute (ITI)
Analog IC Design Track
Summer Camp 2026
Lab 6
Differential Amplifier
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PART 1: Sizing Chart

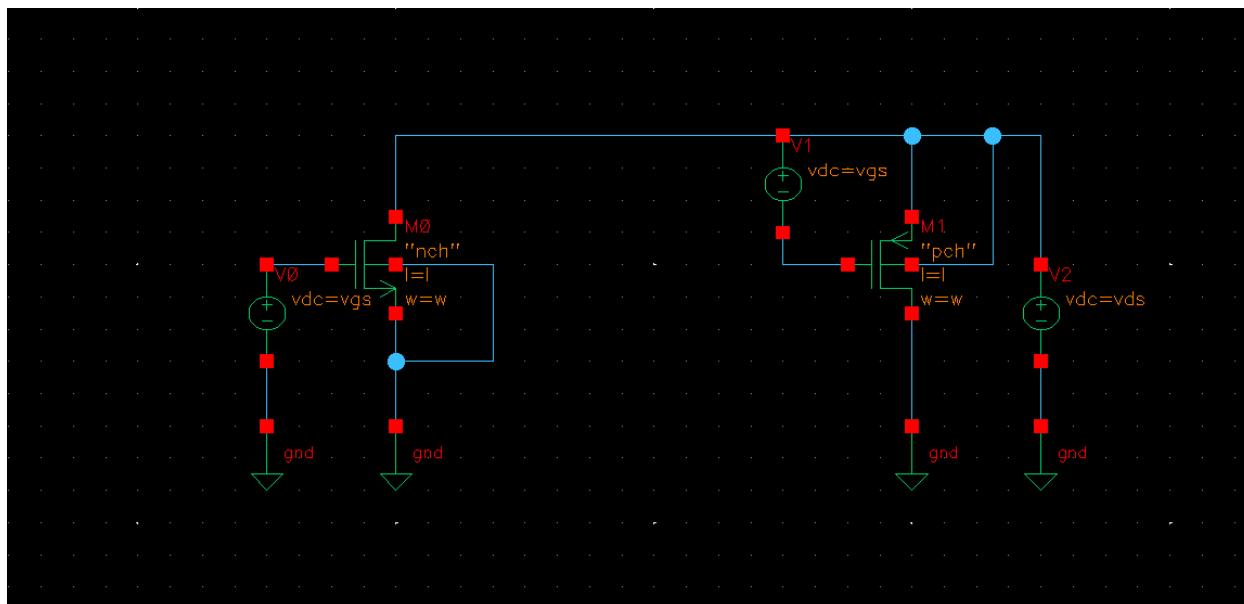


Figure 1:sizing schemtic

1) Design Specifications and Initial Calculations

For this design, the 180 nm technology node is utilized with a channel length of $L = 1 \mu\text{m}$. The target specifications require a differential gain of $|A_v| = 8$ and a common-mode output level of 0.7 V. Given a total tail current of $I_{SS} = 40 \mu\text{A}$, the drain current for each transistor in the differential pair is $I_D = 20 \mu\text{A}$.

Calculation of Load Resistor (R_D): To achieve the specified common-mode output level, the voltage drop across the load resistor (V_{RD}) must be 0.7 V.

Therefore, R_D is calculated as:

$$R_D = \frac{V_{RD}}{I_D} = \frac{0.7}{20 \times 10^{-6}} = 35 \text{ k}\Omega$$

Calculation of V^* : Using the design equation for the differential amplifier's gain, the required V^* parameter is determined to meet the gain specification:

$$|A_v| = \frac{2V_{RD}}{V^*} \text{ \& } V^* = \frac{2 \times 0.7}{8} = 0.175 \text{ V}$$

2) Sizing Chart Extraction (V_{GSQ} and V_{ovQ})

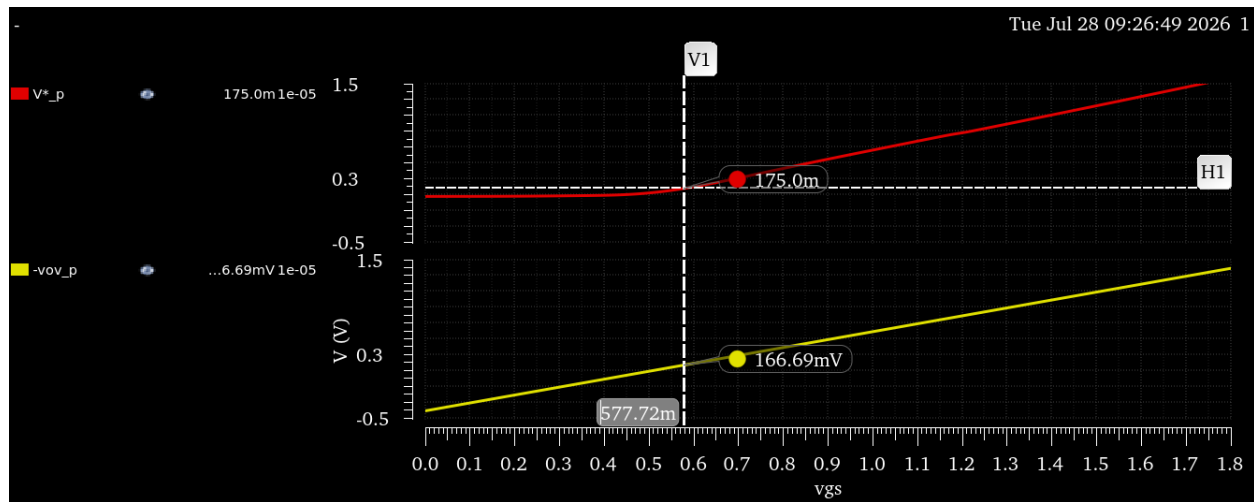


Figure 2: V^* & V_{ov} graphs

Following the calculation of the required target $V^* = 0.175$ V, the device sizing chart was utilized to extract the corresponding operating point parameters. A testbench for a PMOS transistor with an initial width of $W = 10 \mu\text{m}$ was simulated, sweeping V_{GS} to plot both V^* and V_{ov} overlaid.

By locating the calculated V^* value on the characteristic curve (as shown in the figure above), the required quiescent gate-source voltage (V_{GSQ}) and overdrive voltage (V_{ovQ}) were directly extracted from the simulation markers:

Target V^* : 175.0 mV

Extracted V_{GSQ} : 577.72 mV

Extracted V_{ovQ} : 166.69 mV

3) Final Device Sizing (Width Calculation)

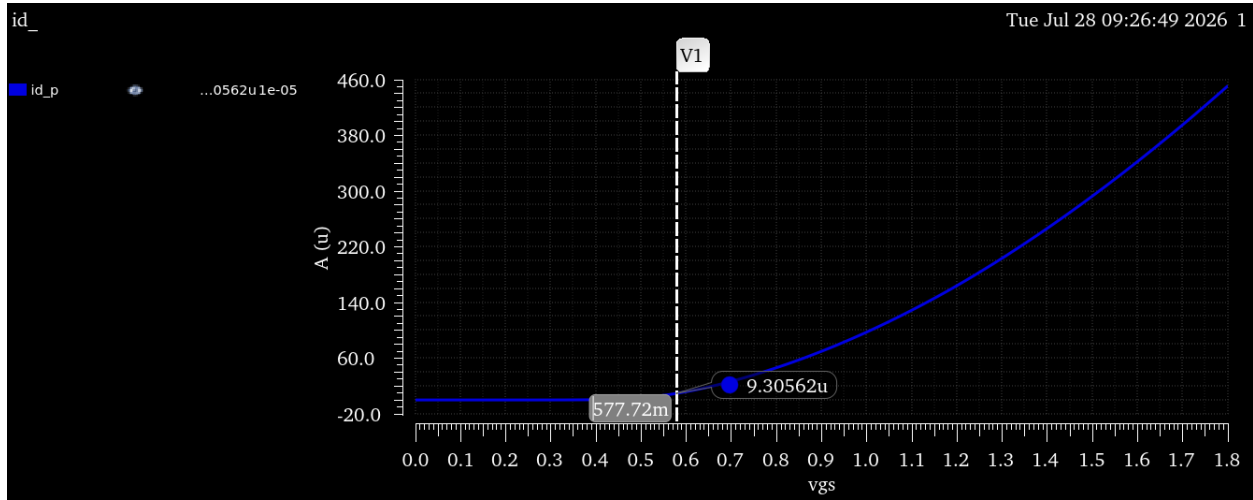


Figure 3: I_{DX} at $W=10 \mu\text{m}$

With the operating point established at $V_{GSQ} = 577.72 \text{ mV}$, the corresponding drain current (I_{DX}) for the reference device ($W = 10 \mu\text{m}$) was extracted from the I_D vs. V_{GS} characteristic curve. As shown in the plotted simulation, the reference current is measured as $I_{DX} = 9.30562 \mu\text{A}$.

Since the drain current is directly proportional to the transistor width irrespective of the operating region, cross-multiplication (ratio and proportion) is applied to determine the actual width (W) required to carry the target bias current of $I_{DQ} = 20 \mu\text{A}$:

$$W = W_{\text{ref}} \times \frac{I_{DQ}}{I_{DX}}$$

$$W = 10 \mu\text{m} \times \frac{20 \mu\text{A}}{9.30562 \mu\text{A}} \approx 21.49 \mu\text{m}$$

Therefore, the final synthesized width for each PMOS transistor in the differential pair is set to $21.49 \mu\text{m}$.

PART 2: Differential Amplifier

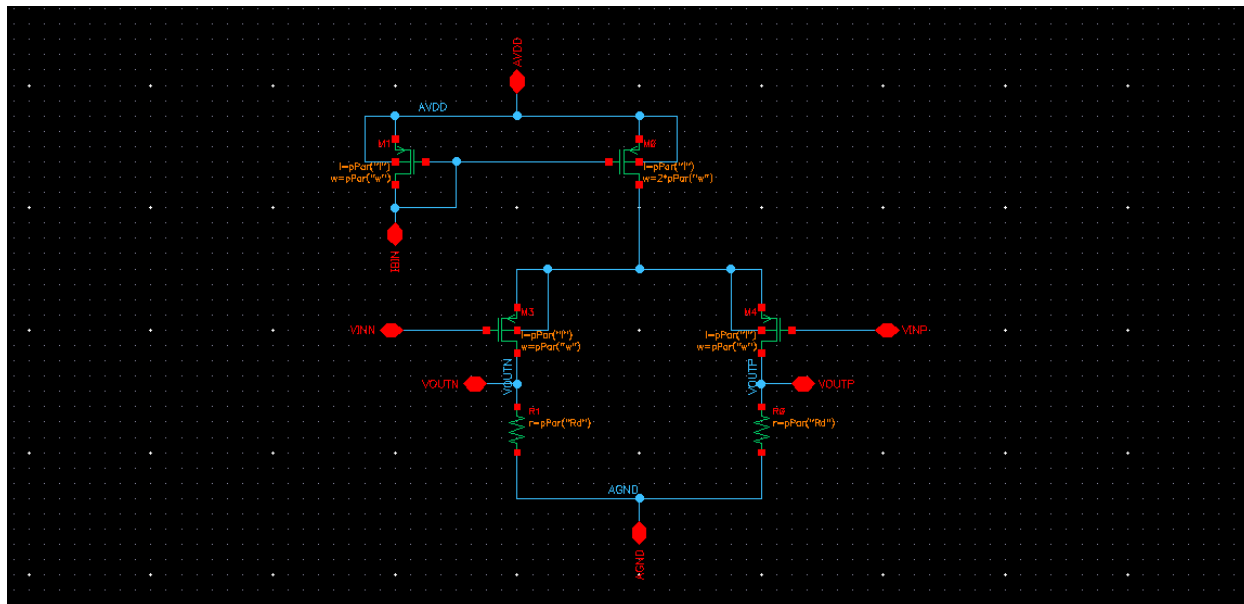


Figure 4: diff. amp schemtic

Following the device sizing and operating point extraction, the complete schematic of the differential amplifier was constructed as shown in the figure above. The circuit utilizes a PMOS input differential pair loaded with the previously calculated resistors (R_D).

Current Mirror Biasing (2:1 Ratio): To provide the required total tail current ($I_{SS} = 40 \mu\text{A}$) specified for the amplifier, a simple current mirror topology is implemented for the bias current source. As defined in the lab specifications, an input reference bias current of $I_{BIN} = 20 \mu\text{A}$ is used.

To achieve the required $40\text{ }\mu\text{A}$ tail current, the current mirror is designed with a **2:1 mirroring ratio**. This is explicitly implemented in the schematic by setting the width of the tail current source transistor (M2) to be twice the width of the reference diode-connected transistor (M1) ($W_{M2} = 2 \times W_{M1}$). Consequently, the tail current splits equally between the two branches of the fully balanced differential pair, providing exactly $I_D = 20\text{ }\mu\text{A}$ for each input PMOS transistor, which completely satisfies the initial design constraints.

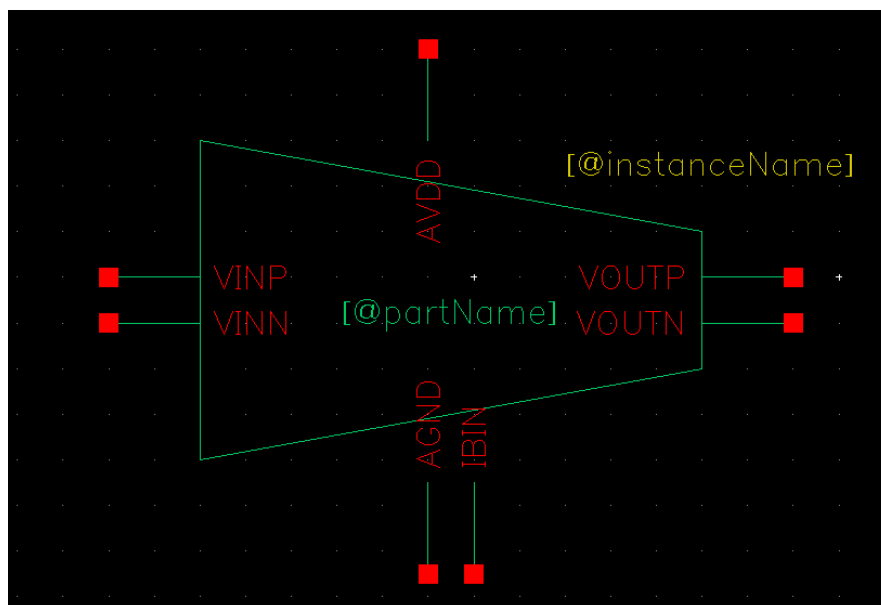


Figure 5:diff. amp symbol

Testbench Setup for Differential and Common-Mode Analysis

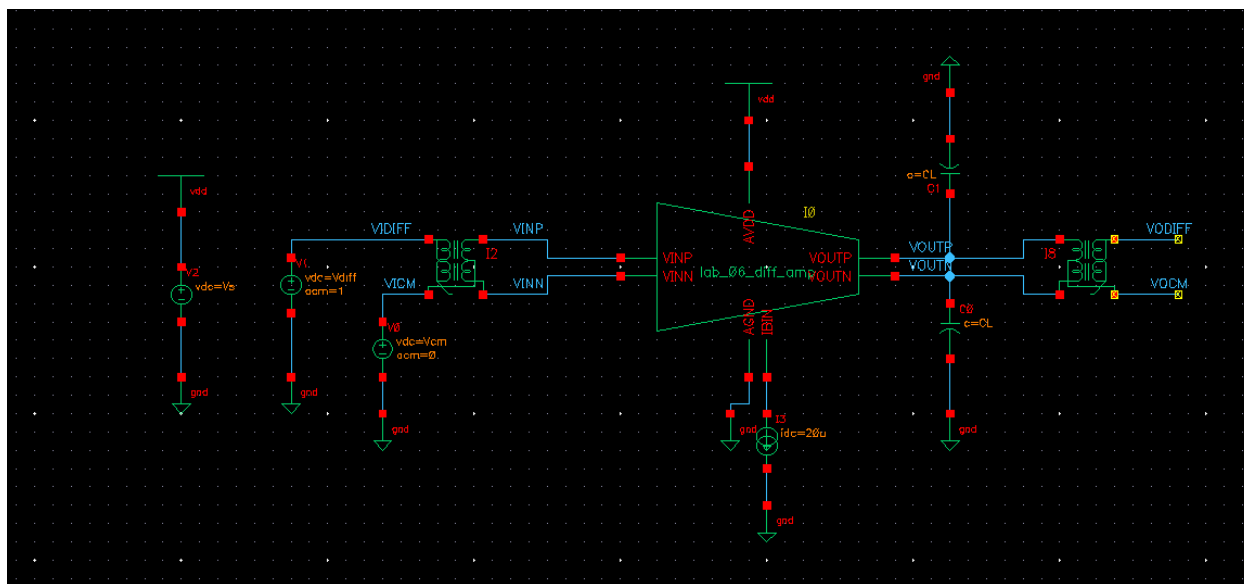


Figure 6:tb_circuit



To evaluate the small-signal and large-signal characteristics of the designed amplifier, the testbench schematic was constructed as shown in the figure above.

To facilitate the independent application and measurement of differential and common-mode signals, the ideal_balun component from the analogLib library is utilized at both the input and output stages. The input balun decomposes the common-mode (V_{ICM}) and differential (V_{IDIFF}) voltage sources into the respective gate signals (V_{INP} and V_{INN}). Similarly, the output balun combines the single-ended outputs (V_{OUTP} and V_{OUTN}) to extract the output differential (V_{ODIFF}) and common-mode (V_{OCM}) voltages.

Furthermore, load capacitors of $C_L = 1$ pF are connected to the output nodes as specified. A DC current source of $20 \mu\text{A}$ is connected to the I_{BIN} pin, which drives the internal current mirror to properly bias the amplifier. This setup allows for seamless configuration of the AC magnitudes depending on whether a differential or common-mode analysis is being performed.

1) OP simulation.

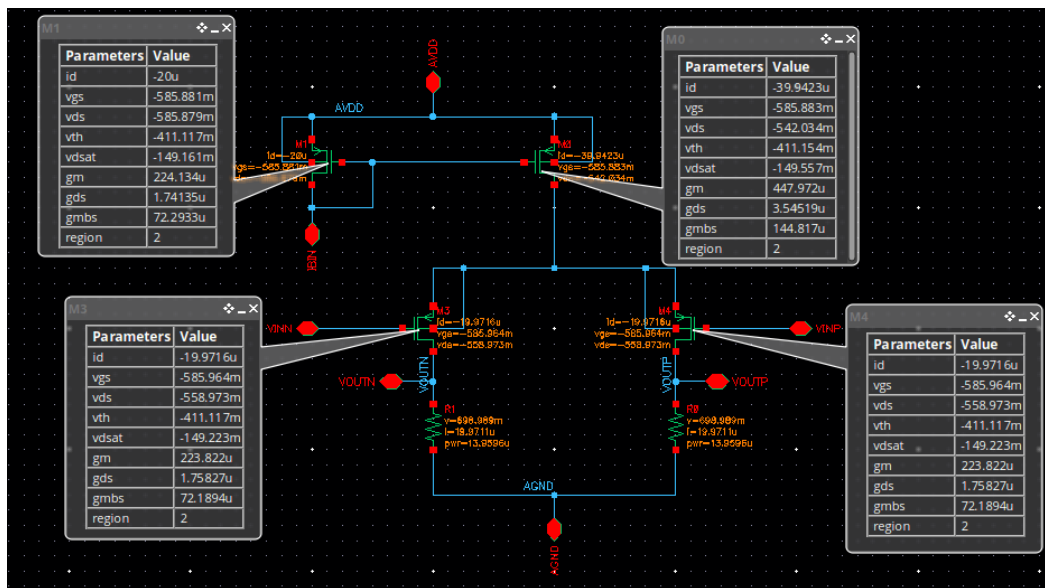


Figure 7:DC_OP



A DC operating point (DC OP) simulation was performed to verify the biasing conditions and ensure all transistors are operating in the proper region. The annotated schematic above confirms the successful implementation of the design specifications.

Current Mirror and Biasing:

The reference transistor (M1) is driven by the ideal input bias current of **-20 μA** . The 2:1 current mirror successfully scales this current, providing a tail current of **-39.94 μA** through the tail current source (M0). The slight deviation from the ideal **40 μA** is expected and is attributed to channel length modulation, as the drain-to-source voltages (V_{ds}) of M1 and M0 are not perfectly identical.

Differential Pair Operation:

The tail current is evenly split between the symmetrical branches of the differential pair. Transistors M3 and M4 each draw approximately **-19.97 μA** , which closely matches the target **-20 μA** specified in the initial design requirements.

Operating Region and Transconductance:

Crucially, the operating point parameters confirm that all PMOS devices (M1, M0, M3, and M4) are operating in the active saturation region (region = 2), which is essential for linear amplification. Furthermore, the extracted transconductance for the input differential pair is $g_m = 223.8 \mu\text{S}$. This parameter is critical and will be used to calculate the simulated differential voltage gain ($A_v = g_m R_D$) to compare it against the theoretical target.

2) Diff small signal ccs:

To evaluate the small-signal differential characteristics, an AC analysis was performed over a frequency range of 1 Hz to 10 GHz. The AC magnitude of the differential source was set to 1, while the common-mode AC source was disabled (set to 0). The DC common-mode input voltage (V_{ICM}) was biased at the center of the previously extracted CMIR.

High-Frequency Roll-off Analysis:

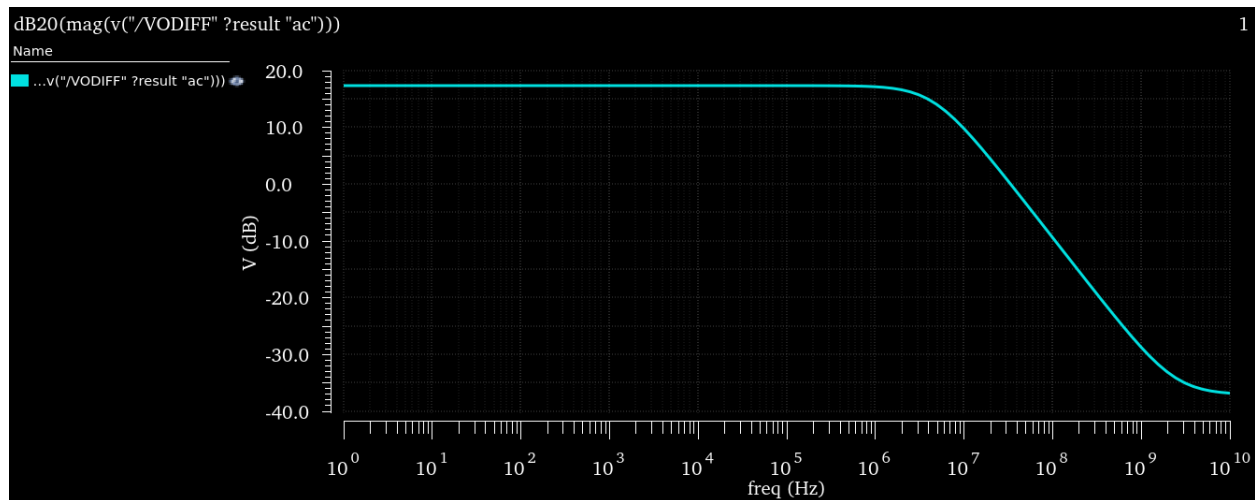


Figure 8: AC analysis for diff. input

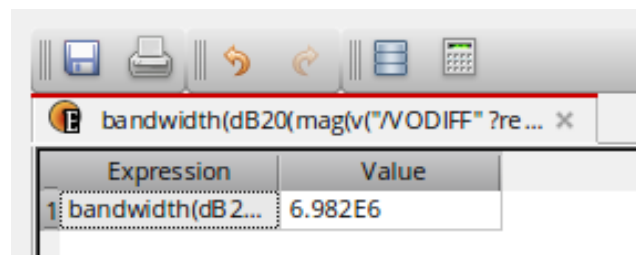


Figure 9: simulation Bandwidth

As illustrated in the Bode plot above, the amplifier maintains a constant differential gain at low frequencies (approximately 17.36 dB, which equates to 7.38 V/V). However, as the operating frequency increases, the gain experiences a significant roll-off.

This high-frequency degradation is primarily dictated by the dominant pole at the output node. At elevated frequencies, the explicit load capacitance ($C_L = 1$ pF) along with the MOSFET's internal parasitic capacitances (specifically the drain-to-

bulk capacitance C_{db} and the gate-to-drain capacitance C_{gd}) act as a low-impedance bypass path to AC ground. This capacitive shunting effect reduces the effective load impedance, thereby decreasing the voltage gain according to the dominant pole frequency equation :

$$\omega_p \approx \frac{1}{R_D(C_L + C_{db} + C_{gd})} = \frac{1}{35k \times 1p} = 2\pi \times (4.547 \text{ MHz})$$

Comparison with Hand Analysis: The table below summarizes the comparison between the theoretical hand calculations and the extracted simulation results for the differential small-signal performance.

Parameter	Theoretical (Hand Analysis)	Simulated (AC Analysis)
DC Differential Gain (A_{vd})	7.833V/V 17.879 dB	7.38 V/V 17.36 dB
Bandwidth (BW / f_{3dB})	4.547 z	6.982 z

3) Common-Mode Small-Signal Characteristics and CMRR

To evaluate the common-mode rejection capabilities of the amplifier, an AC analysis was performed with the common-mode AC magnitude set to 1 and the differential AC source disabled.

Common-Mode DC Gain (A_{vcm}) and Frequency Response:

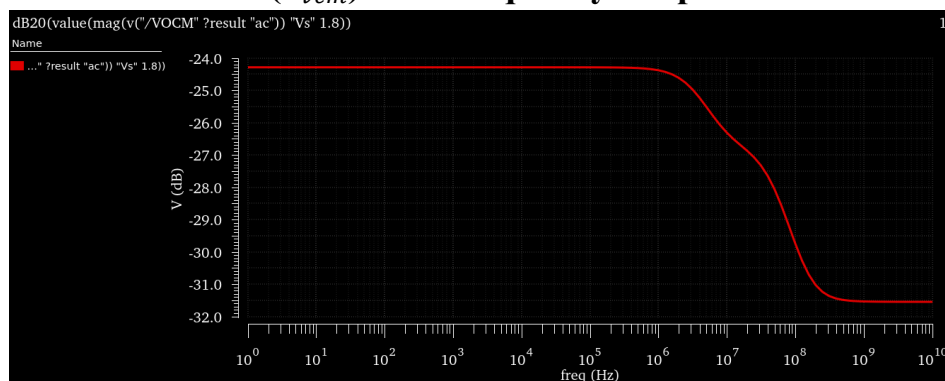


Figure 10: A_{vcm}

As shown in the common-mode Bode plot, the DC common-mode gain is approximately **-24.2 dB** (which corresponds to a linear gain of **0.061 V/V**). This value is significantly smaller than 1. Theoretically, this is expected because the common-mode gain is heavily degenerated by the high output resistance (r_{o0}) of the tail current source (M0). The hand analysis formula confirms this behavior:

$$A_{vcm} \approx \frac{-R_D}{2r_{o0}}$$

Since $2r_{o0} \gg R_D$, the resulting gain is highly attenuated, providing good common-mode rejection.

At high frequencies, the common-mode gain experiences a roll-off (dropping to roughly -31.5 dB). This variation is caused by the parasitic capacitances at the tail node (the common-source node of the differential pair, primarily the drain-to-bulk capacitance C_{db0} of M0 and the source-to-bulk capacitances C_{sb} of M3/M4). At high frequencies, these parasitic capacitors shunt the large tail resistance r_{o0} , introducing a zero that reduces the effective tail impedance. However, the dominant pole formed by the load capacitance (C_L) and R_D at the output node eventually dominates, causing the overall common-mode gain to roll off.

CMRR (A_{vd}/A_{vcm}) vs. Frequency:

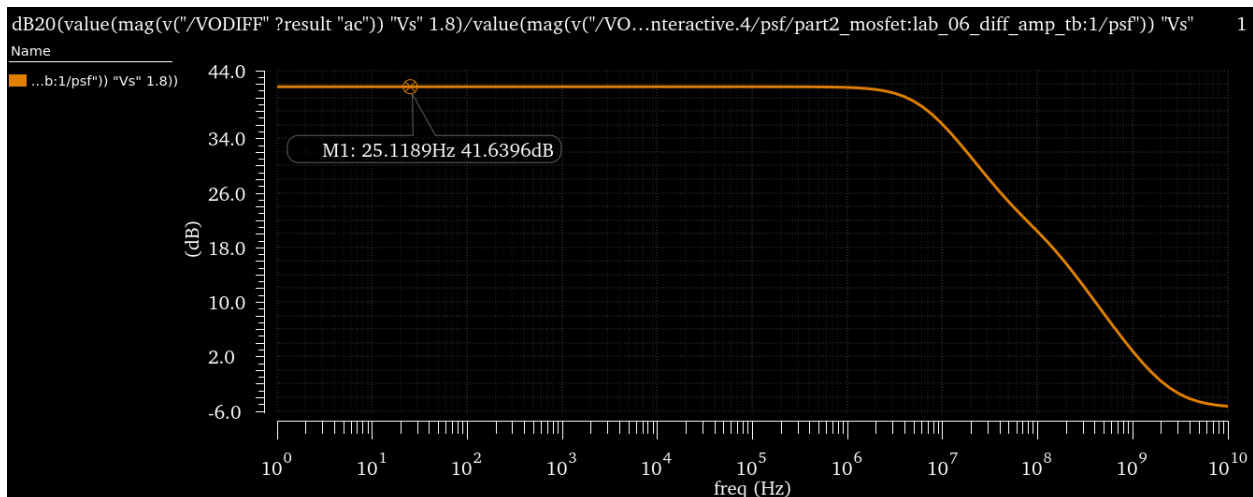


Figure 11: (CMRR) vs. frequency.



The Common-Mode Rejection Ratio (CMRR) defines the amplifier's ability to amplify differential signals while rejecting common-mode noise. From the simulation, the DC CMRR is exceptionally high at **41.64 dB**.

However, as the frequency increases, the CMRR degrades significantly. This variation is justified because the differential gain (A_{vd}) rolls off at a lower frequency due to the output pole (R_D and C_L), while the common-mode gain (A_{vcm}) degradation is partially compensated by the zero introduced by the tail capacitance. Since A_{vd} decreases faster than A_{vcm} at high frequencies, their ratio (CMRR) inherently drops.

4) Differential Large-Signal Characteristics

To investigate the large-signal differential behavior of the amplifier, a DC sweep was performed on the differential input voltage (V_{id}) from -1.8 V to $+1.8$ V ($-V_{DD}$ to $+V_{DD}$), while maintaining the common-mode input voltage (V_{ICM}) at the center of the previously determined CMIR.

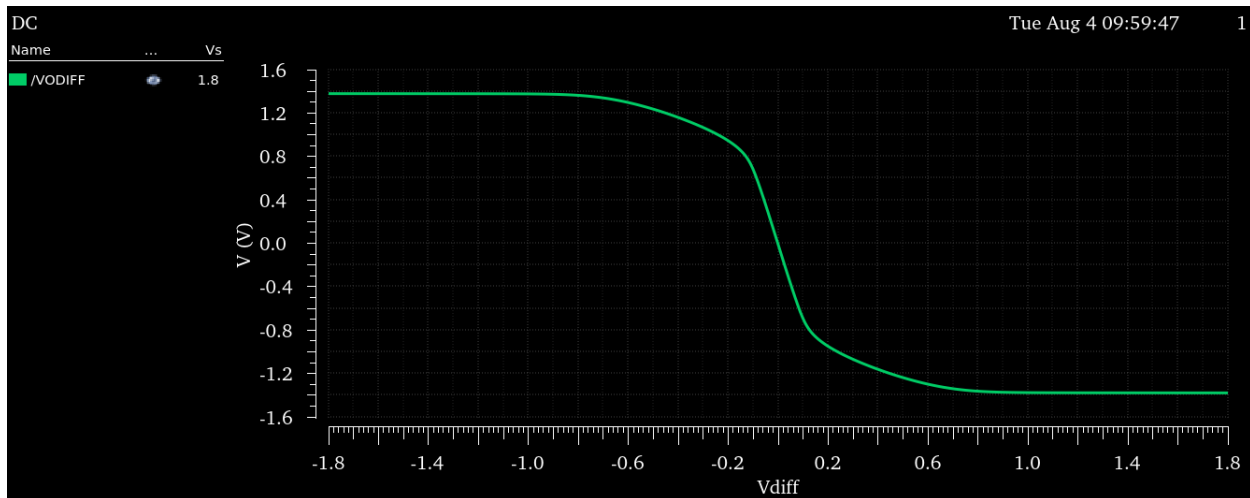


Figure 12: V_{ODIFF} vs. V_{id}

As depicted in the transfer characteristic curve above, the amplifier exhibits linear operation for small differential inputs. However, for large input swings, the tail current ($I_{SS} = 40 \mu\text{A}$) completely steers to one of the differential branches.



Extreme Values Hand Analysis:

When the differential input is completely unbalanced (i.e., very large V_{id}), one input transistor turns fully ON, absorbing the entire tail current I_{SS} , while the other transistor completely cuts off.

Therefore, the maximum single-ended output voltage drop is $I_{SS} \times R_D$, and the other is 0 V.

The extreme limits for the differential output voltage (V_{ODIFF}) can be theoretically calculated as:

$$V_{ODIFF(max,min)} = \pm I_{SS} \times R_D = \pm(40 \mu\text{A} \times 35 \text{ k}\Omega) = \pm 1.4 \text{ V}$$

The table below demonstrates the perfect alignment between the theoretical hand calculations and the saturated extreme values extracted from the simulation plot.

Parameter	Theoretical Extreme Value	Simulated Extreme Value
Max Positive Output ($+V_{ODIFF}$)	+1.4 V	+1.4 V
Max Negative Output ($-V_{ODIFF}$)	-1.4 V	-1.4 V

5) CM large signal ccs (region vs V_{ICM}):

To determine the Common-Mode Input Range (CMIR), a large-signal DC sweep was performed on the common-mode input voltage (V_{ICM}) from 0 V to 1.8 V (V_{DD}). The operating region parameter of the transistors was plotted to identify the range where all critical devices remain in saturation.

• Region Parameter Method

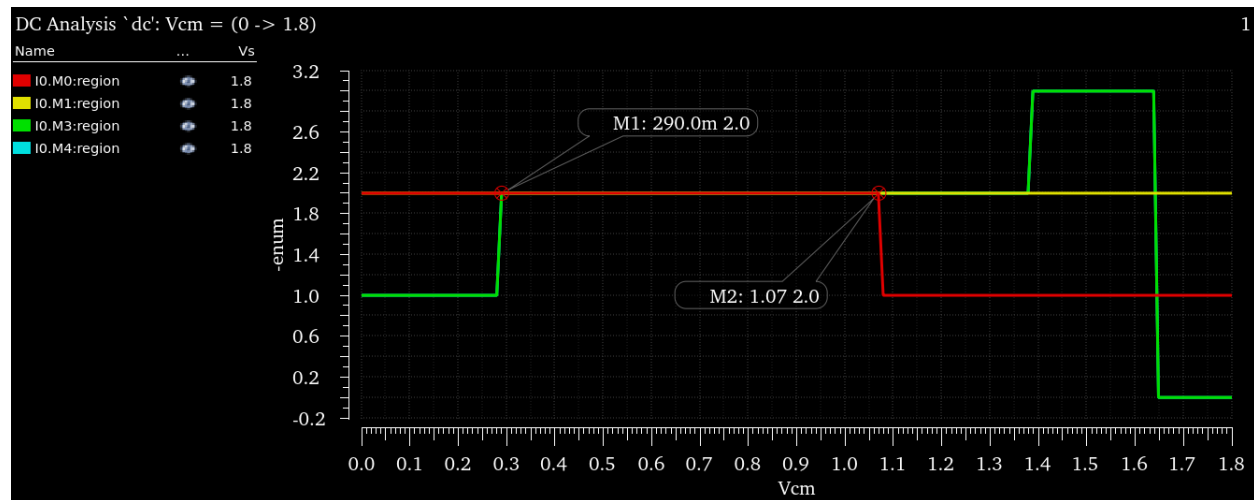


Figure 13: regions of the tail current source (M0) and input pair (M3/M4) vs. V_{ICM}

For proper differential amplifier operation, both the input differential pair (M3 and M4) and the tail current source (M0) must operate in the saturation region (region = 2). Based on the simulation results:

- **Lower Bound ($V_{in(min)}$):** The input pair transistors (M3 and M4) transition from the triode region (region = 1) into the saturation region (region = 2) at $V_{ICM} = 0.29$ V.
- **Upper Bound ($V_{in(max)}$):** The tail current source (M0) is forced out of saturation and enters the triode region when V_{ICM} exceeds 1.07 V.

Therefore, the simulated Common-Mode Input Range based on the region parameter is:

$$0.29 \text{ V} \leq V_{ICM} \leq 1.07 \text{ V}$$



Methodology Drawback: It is important to note that the drawback of this method is that the "region" parameter is a simulator-specific variable that cannot be experimentally measured in a physical laboratory. Furthermore, it acts as an abrupt mathematical boundary and is not quantitatively related to gradual circuit performance degradations (such as gain reduction or distortion).

Theoretical Hand Analysis for CMIR

To validate the simulated CMIR, theoretical hand calculations were performed using the absolute values of the extracted DC operating point parameters ($|V_{THp}| = 0.411 \text{ V}$, $|V_{GS3}| = 0.586 \text{ V}$, and tail saturation voltage $|V_{DSAT0}| \approx 0.150 \text{ V}$).

1. Minimum Common-Mode Input Voltage ($V_{in(min)}$):

The lower limit is determined by the condition to keep the input differential pair (M3/M4) in saturation. The gate voltage must not drop significantly below the drain voltage:

$$V_{in(min)} \geq V_D - |V_{THp}|$$

Since $V_D = I_D \times R_D = 20 \mu\text{A} \times 35 \text{ k}\Omega = 0.7 \text{ V}$:

$$V_{in(min)} = 0.7 \text{ V} - 0.411 \text{ V} = 0.289 \text{ V}$$

2. Maximum Common-Mode Input Voltage ($V_{in(max)}$):

The upper limit is restricted by the tail current source (M0). To keep M0 in saturation, the common-source node (tail voltage) must be sufficiently below V_{DD} :

$$V_{tail(max)} = V_{DD} - |V_{DSAT0}|$$

Relating this to the input gate voltage:

$$V_{in(max)} = V_{DD} - |V_{DSAT0}| - |V_{GS3}|$$

$$V_{in(max)} = 1.8 \text{ V} - 0.150 \text{ V} - 0.586 \text{ V} = 1.064 \text{ V}$$



Comparison Table:

Parameter	Theoretical Hand Analysis	Simulated (Region Method)
$V_{in(min)}$	0.289 V	0.29 V
$V_{in(max)}$	1.064 V	1.07 V

6) Gain-Drop Method for CMIR Extraction (GBW vs. V_{ICM})

A more practical and spec-driven method to determine the CMIR is to observe the degradation of the differential gain (A_{vd}) as the common-mode input (V_{ICM}) varies. Since the bandwidth is primarily set by R_D and C_L and remains relatively constant, the variation in the low-frequency DC gain directly reflects the variation in the Gain-Bandwidth Product (GBW) of the amplifier.

To implement this, an AC analysis was performed at a low frequency (1 Hz) to extract the DC gain, while simultaneously sweeping V_{ICM} from 0 V to 1.8 V using the AC sweep feature over a design variable.

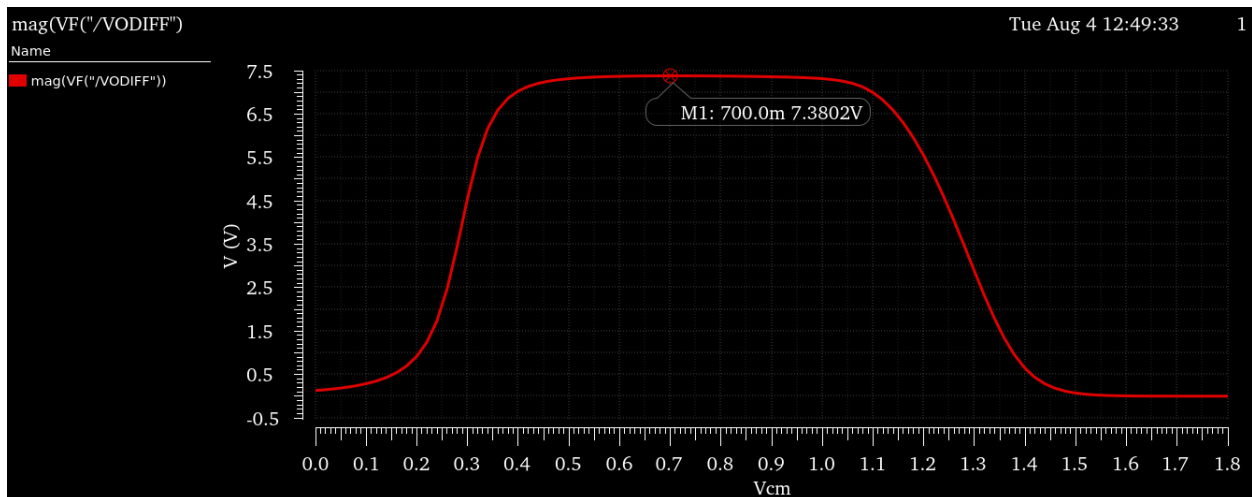


Figure 14: differential gain magnitude A_{vd} vs. V_{ICM}



As shown in the simulated curve, the maximum mid-range differential gain is extracted as **7.38 V/V**. According to the lab specifications, the valid Common-Mode Input Range is defined as the region where the gain remains within 90% of its maximum value (i.e., a maximum allowable gain drop of 10%).

Threshold Calculation:

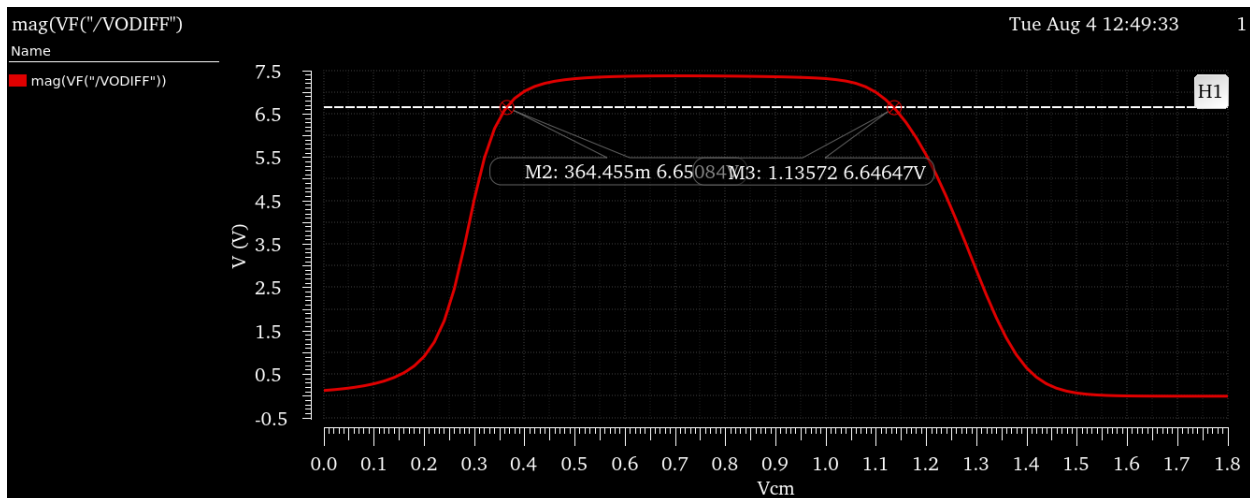
- Maximum Gain ($A_{v(max)}$) = 7.38 V/V
- 90% Threshold Limit = $0.90 \times 7.38 \text{ V/V} = 6.642 \text{ V/V}$

By locating the points on the curve where the gain drops to 6.642 V/V, the bounds of the valid CMIR are identified:

- **Lower Bound ($V_{in(min)}$):** The gain reaches the threshold at $V_{ICM} = 0.364 \text{ V}$.
- **Upper Bound ($V_{in(max)}$):** The gain drops below the threshold at $V_{ICM} = 1.135 \text{ V}$.

Therefore, the CMIR based on the 10% gain-drop specification is:

$$0.364 \text{ V} \leq V_{ICM} \leq 1.135 \text{ V}$$





- **CMIR Methodology Comparison**

The table below summarizes and compares the extracted CMIR using the two different methodologies discussed above.

Methodology	Lower Bound ($V_{in(min)}$)	Upper Bound ($V_{in(max)}$)	Valid CMIR Range
Region Parameter (Theoretical Saturation)	0.290 V	1.07 V	0.29 V – 1.07 V
Gain-Drop Method (10% GBW Degradation)	0.364 V	1.135 V	0.364 V – 1.135 V

Conclusion: While the region parameter provides a strict theoretical boundary indicating absolute saturation, the gain-drop method provides a narrower but more robust functional specification. It guarantees that the amplifier will not only keep its transistors in saturation but will also maintain its required amplification and GBW performance within the specified range.